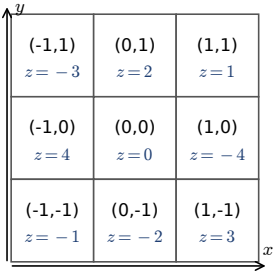
(a) Lo Shu S

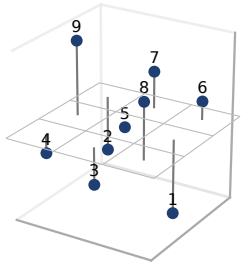
2	7	6
9	5	1
4	3	8

(b) centred cells, $z_{ij} = s_{ij} - 5$

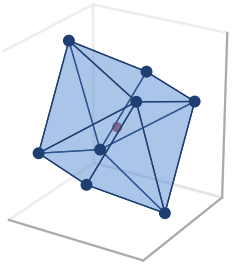
(-1,1) $z=-3$	(0,1) $z=2$	(1,1) $z=1$
(-1,0) $z=4$	(0,0) $z=0$	(1,0) $z=-4$
(-1,-1) $z=-1$	(0,-1) $z=-2$	(1,-1) $z=3$



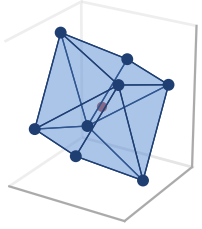
(c) the point set



(d) gem, $f = (8, 16, 10)$

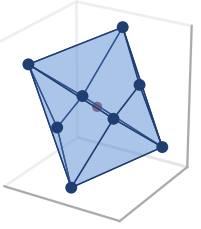


(e) e



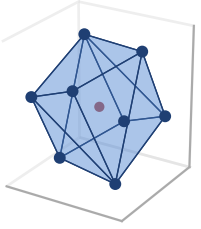
2 7 6 9 5 1 4 3 8

(f) r (rot $90\,^\circ$)



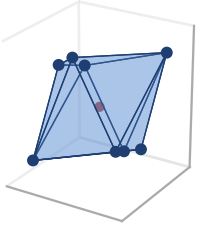
4 9 2 3 5 7 8 1 6

(g) r^2



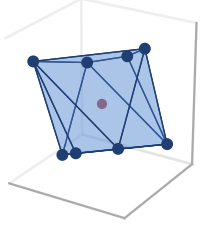
8 3 4 1 5 9 6 7 2

(h) r^3



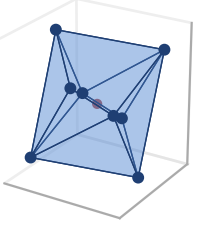
6 1 8 7 5 3 2 9 4

(i) h (left-right)



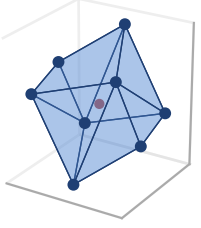
6 7 2 1 5 9 8 3 4

(j) v (top-bottom)



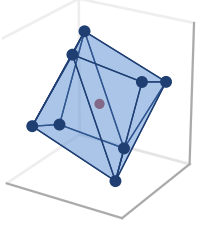
4 3 8 9 5 1 2 7 6

(k) t (transpose)



2 9 4 7 5 3 6 1 8

(l) tr^2 (anti-transpose)



8 1 6 3 5 7 4 9 2